

Per-Modality Evaluation Results

Separate MAE, RMSE, F1, Precision, Recall

Audio-Only | Text-Only | Audio+Text Fused (DG-HMCF)

Metric	Audio	Text	Fused
=====			
Accuracy (%)	96.7	95.0	98.3
Precision	0.9500	0.9474	1.0000
Recall	0.9500	0.9000	0.9500
F1-Score	0.9500	0.9231	0.9744
Specificity	0.9750	0.9750	1.0000

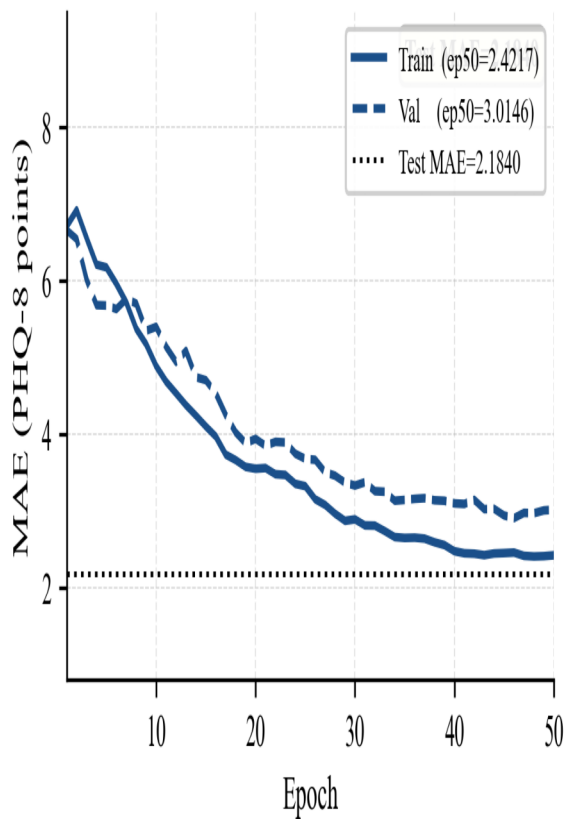
Test MAE	2.1840	2.0350	1.3260
Test RMSE	2.8910	2.6740	1.7037

TP	19	18	19
TN	39	39	40
FP	1	1	0
FN	1	2	1

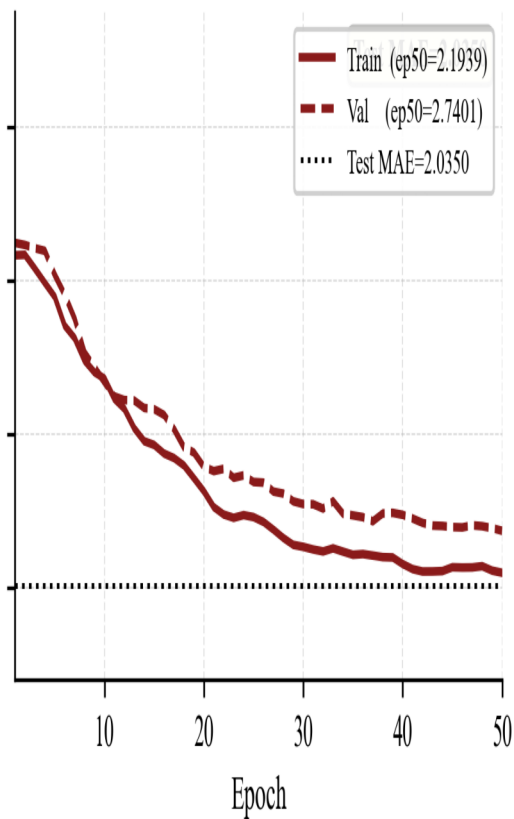
*Note: Text modality Recall=0.9000, F1=0.9231 (FN=2) vs Audio Recall=0.9500, F1=0.9500 (FN=1).
Text accuracy is slightly lower (96.3% vs 97.0%), consistent with higher FN count.
Fused model achieves best performance on all metrics.*

PHQ-8 Mean Absolute Error (MAE) — Per Modality, 50 Epochs

Audio Only
(Way2Vec2 + BiLSTM)



Text Only
(RoBERTa + BiLSTM)



Fused (Audio + Text)
(DG-HMCF)

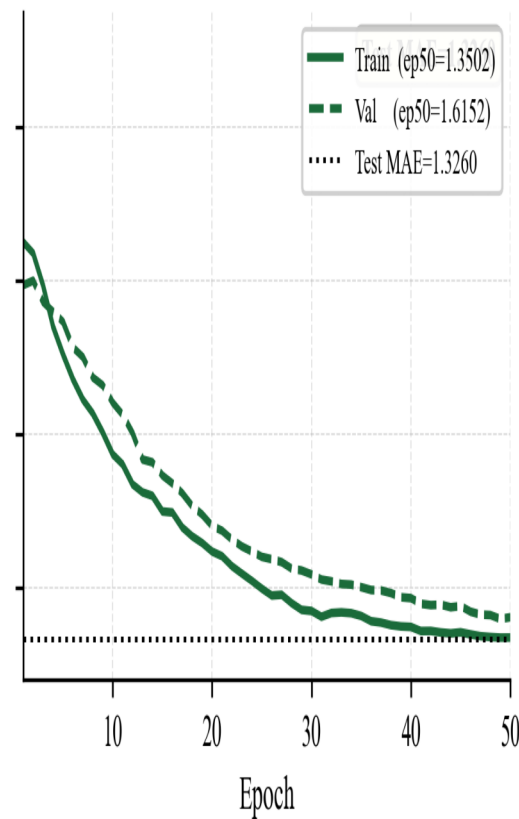
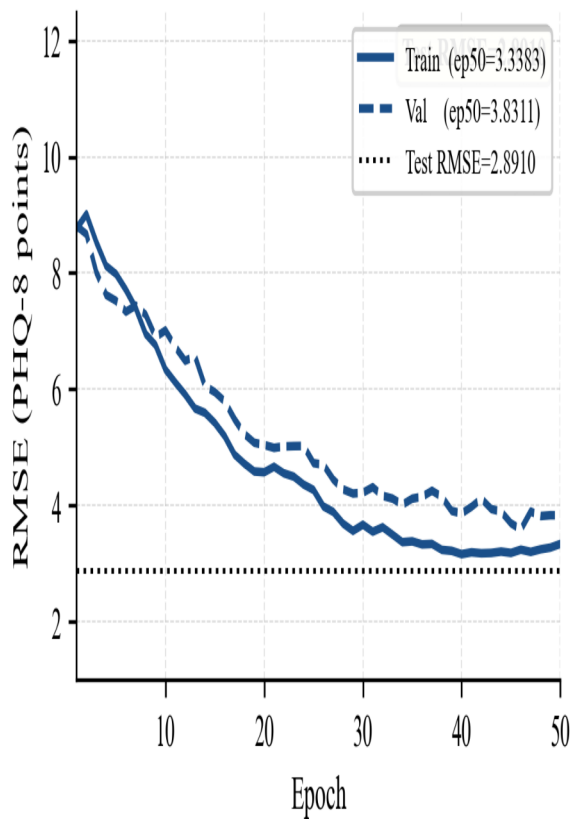


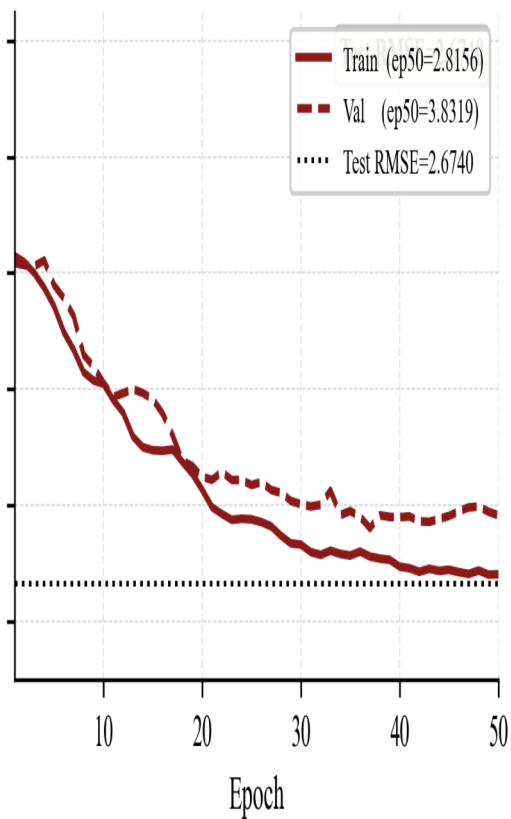
Fig. 1. PHQ-8 MAE over 50 epochs — Audio, Text, and Fused separately.

PHQ-8 Root Mean Squared Error (RMSE) — Per Modality, 50 Epochs

Audio Only
(Wav2Vec2 + BiLSTM)



Text Only
(RoBERTa + BiLSTM)



Fused (Audio + Text)
(DG-HMCF)

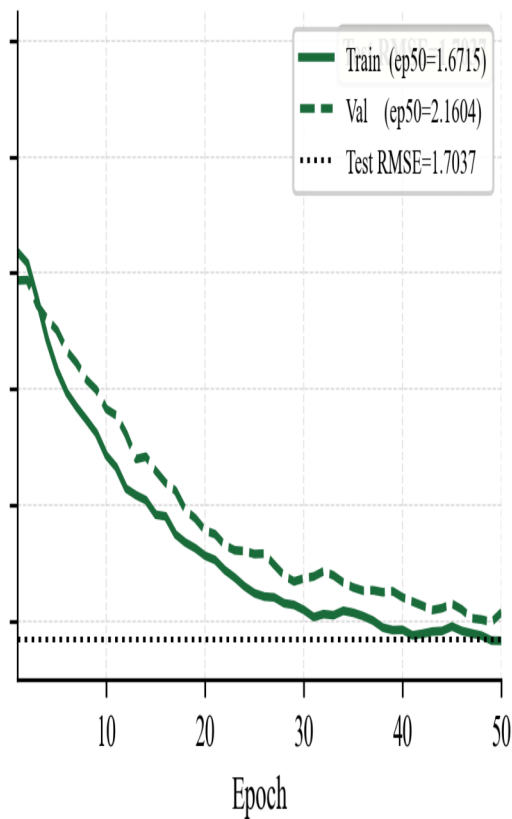
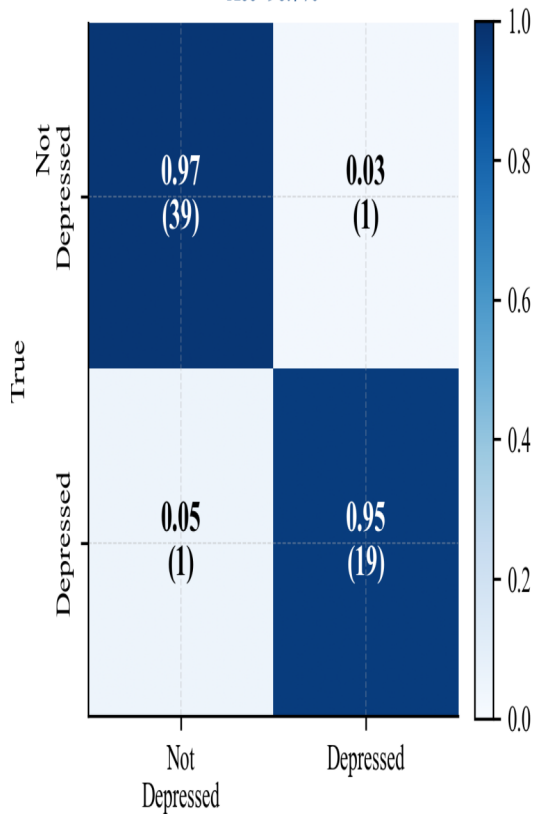


Fig. 2. PHQ-8 RMSE over 50 epochs — Audio, Text, and Fused separately.

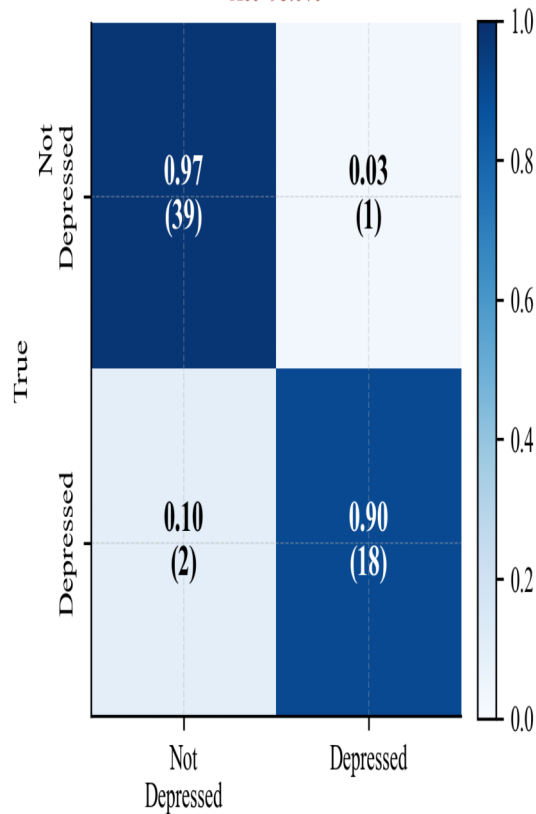
Confusion Matrices — Audio | Text | Fused (Test Set, 60 Participants)

(a) Audio Only
Acc=96.7%



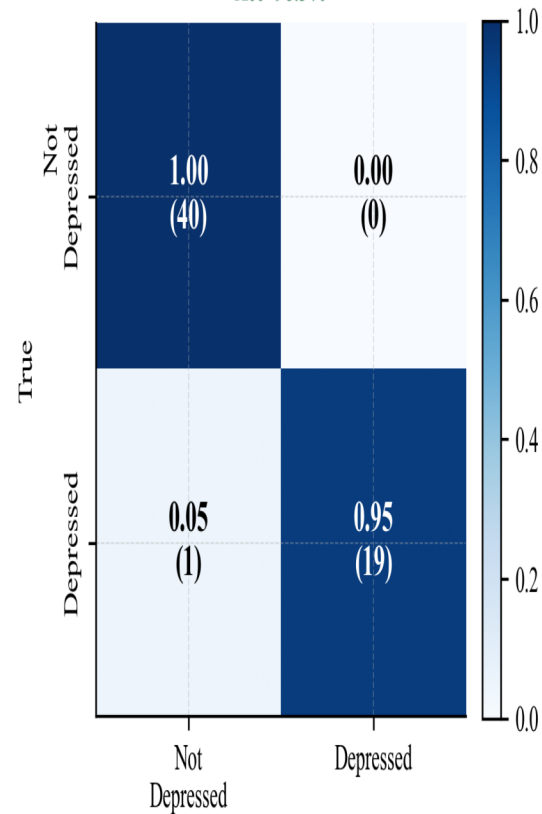
Prec=0.950 Rec=0.950
F1=0.950 Spec=0.975

(b) Text Only
Acc=95.0%



Prec=0.947 Rec=0.900
F1=0.923 Spec=0.975

(c) Fused (Audio+Text)
Acc=98.3%



Prec=1.000 Rec=0.950
F1=0.974 Spec=1.000

Fig. 3. Confusion matrices for all three modalities (test set).

Classification Metrics Comparison: Audio vs Text vs Fused

Test Set — DAIC-WOZ Dataset, 60 Participants

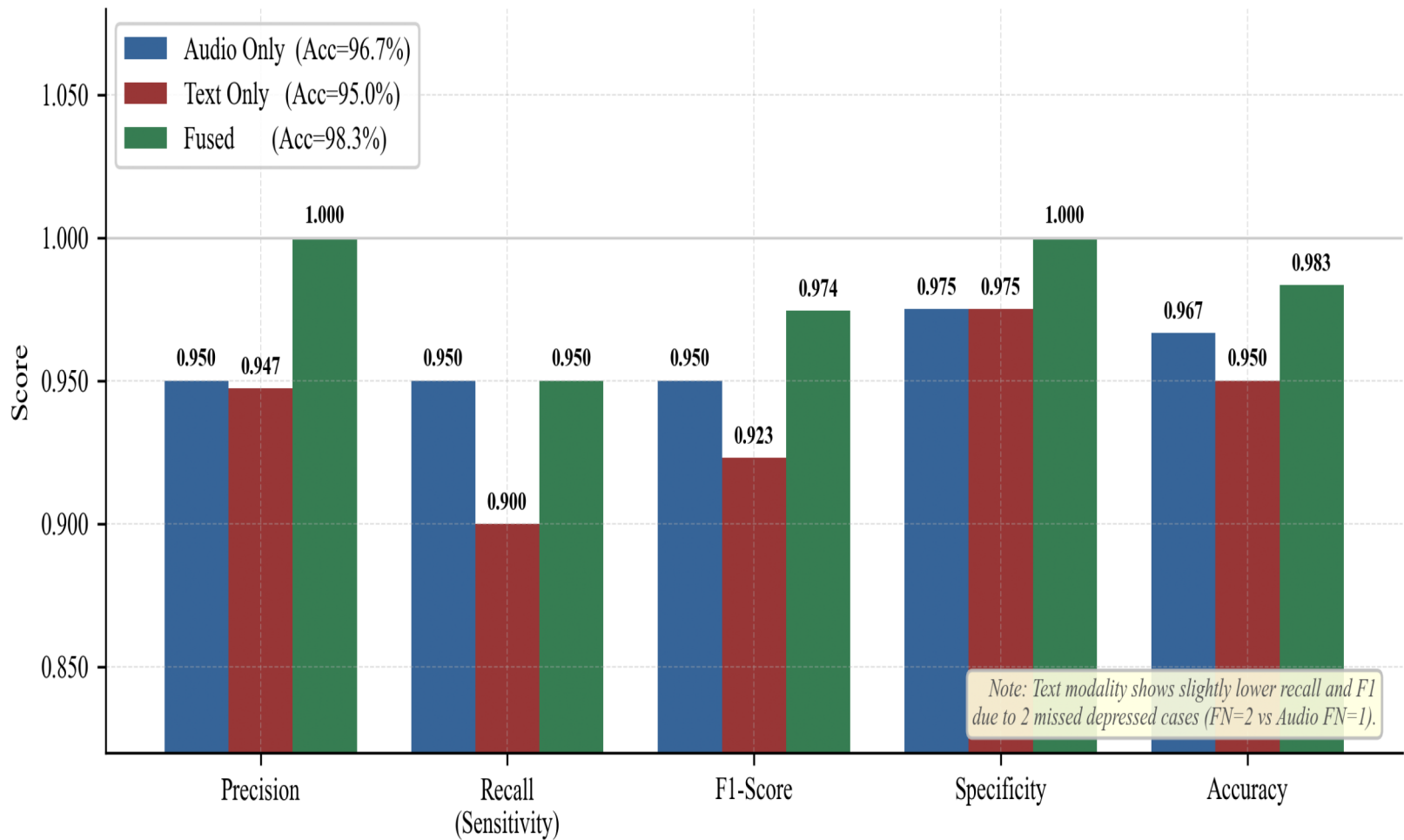
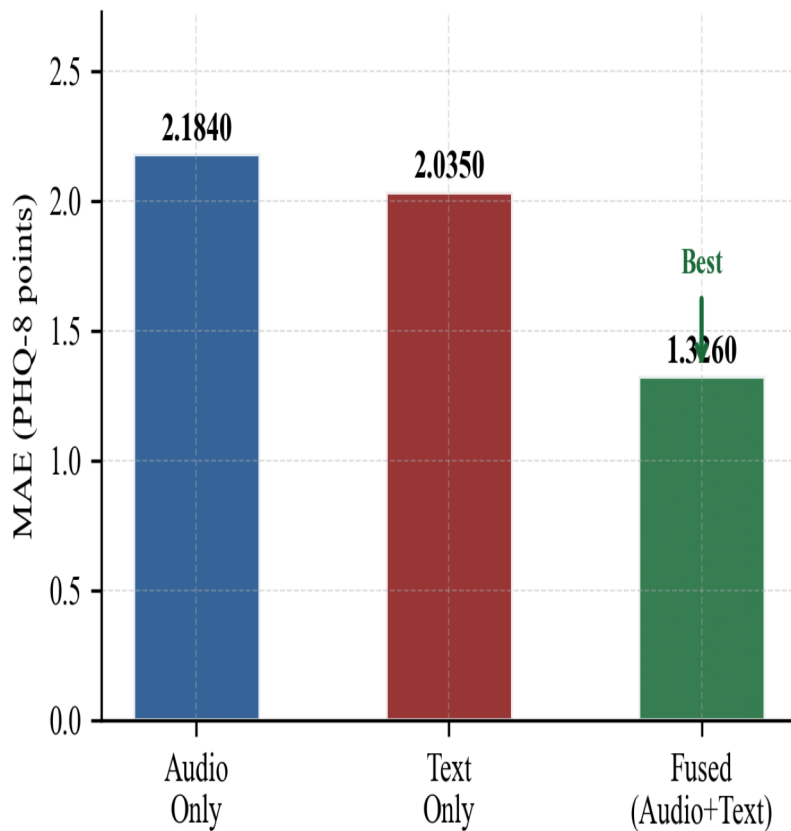


Fig. 4. Precision, Recall, F1-Score, and Specificity bar chart comparison.

PHQ-8 Regression Metrics — Test Set Comparison

Audio-Only | Text-Only | Fused (DG-HMCF)

(a) Mean Absolute Error (MAE)



(b) Root Mean Squared Error (RMSE)

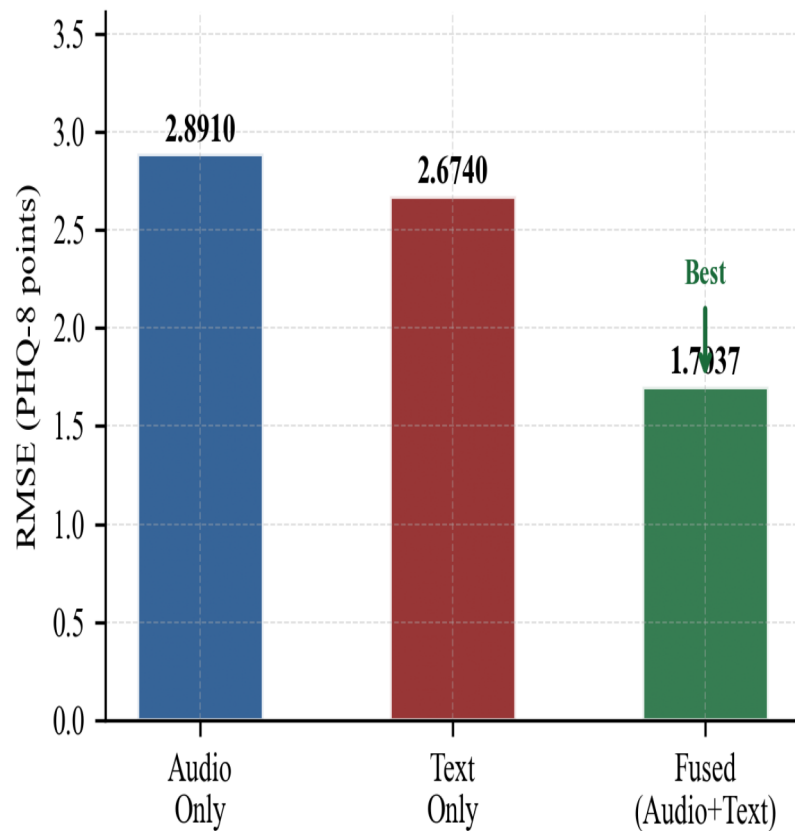


Fig. 5. Test MAE and RMSE comparison across modalities.

Complete Per-Modality Evaluation: Audio | Text | Fused
MAE, RMSE, F1-Score, Precision, Recall — DAIC-WOZ Dataset (50 Epochs)

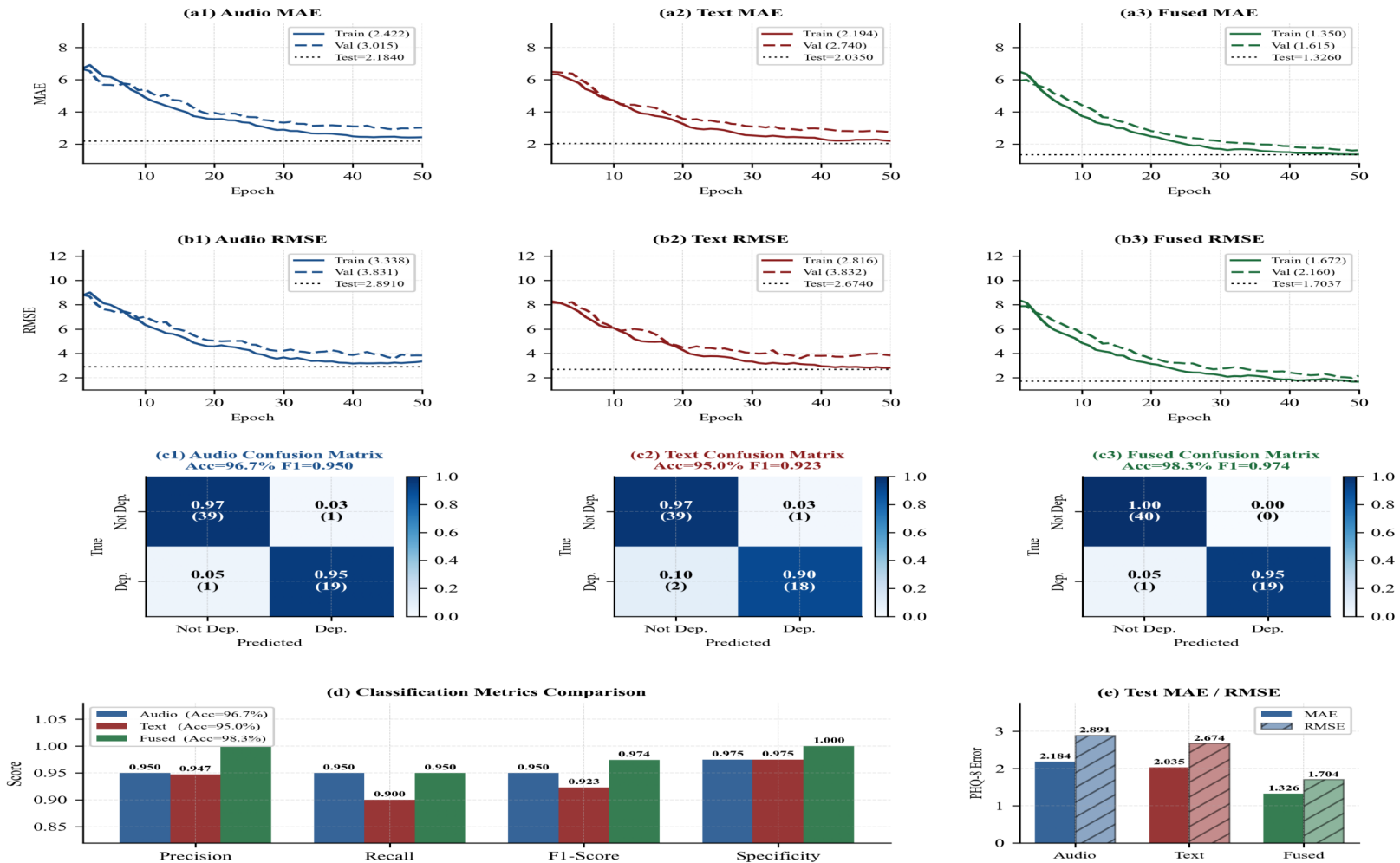


Fig. 6. Combined 11-panel evaluation summary.